SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I A.Sunitha (192511079) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A.Sunitha

Assessment Tasks

Concept Mapping Tasks

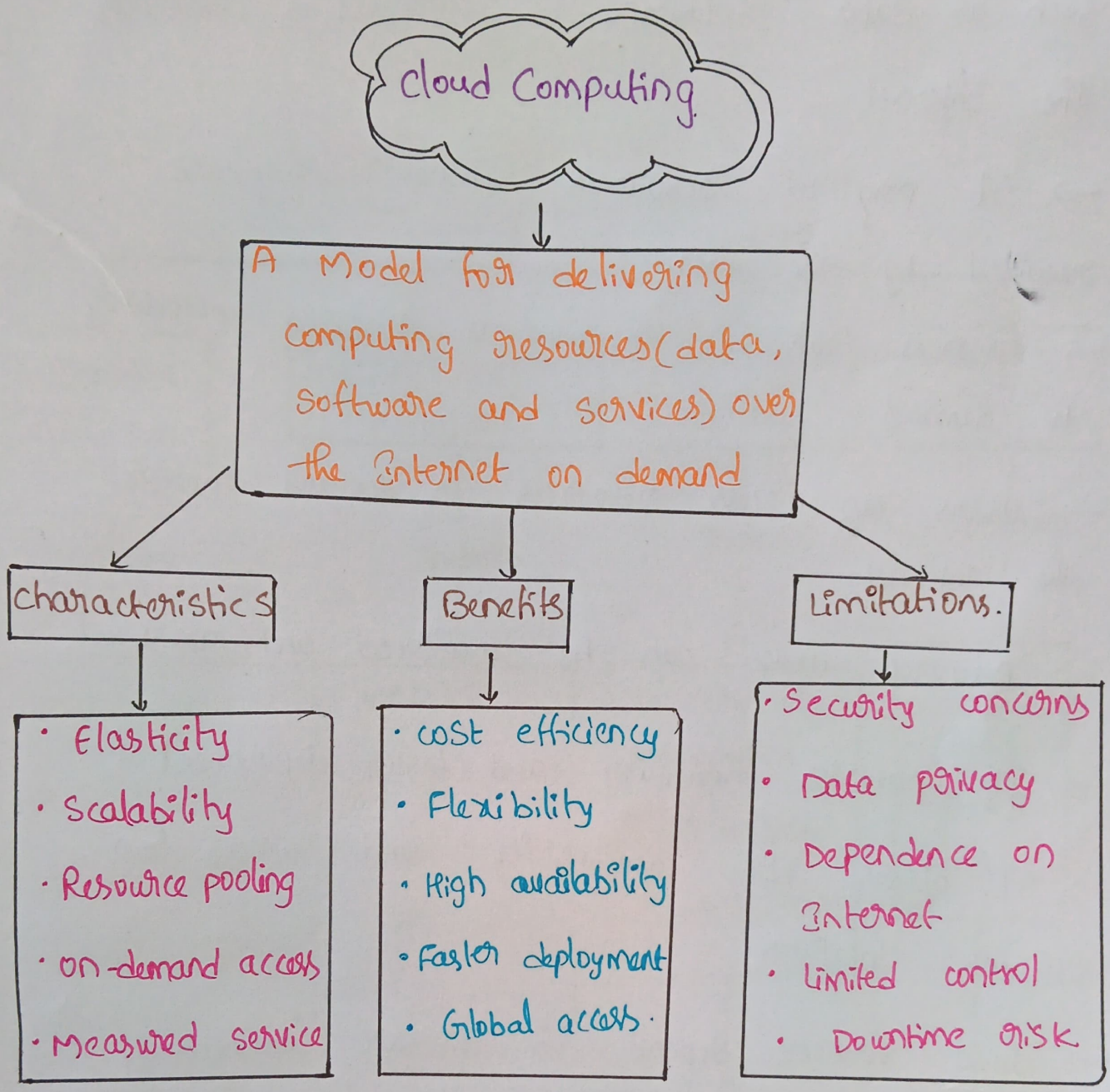
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

1. Cloud computing Fundamentals



Elasticity: Resources can increase or decrease according to demand

Scalability: The system can handle increasing workloads

Resource pooling: Resources are shared among multiple users.

On-Demand Access: Resources are available whenever required.

→ cloud computing delivers computing resources such as data, software and services through the Internet.

→ It provides resources whenever they are required by the user.

→ Resources can be increased or decreased according to demand.

→ users can access resources on demand through the internet.

→ Resource usage can be monitored and measured.

→ It provides flexibility and faster deployment.

→ It offers high availability and access from different locations.

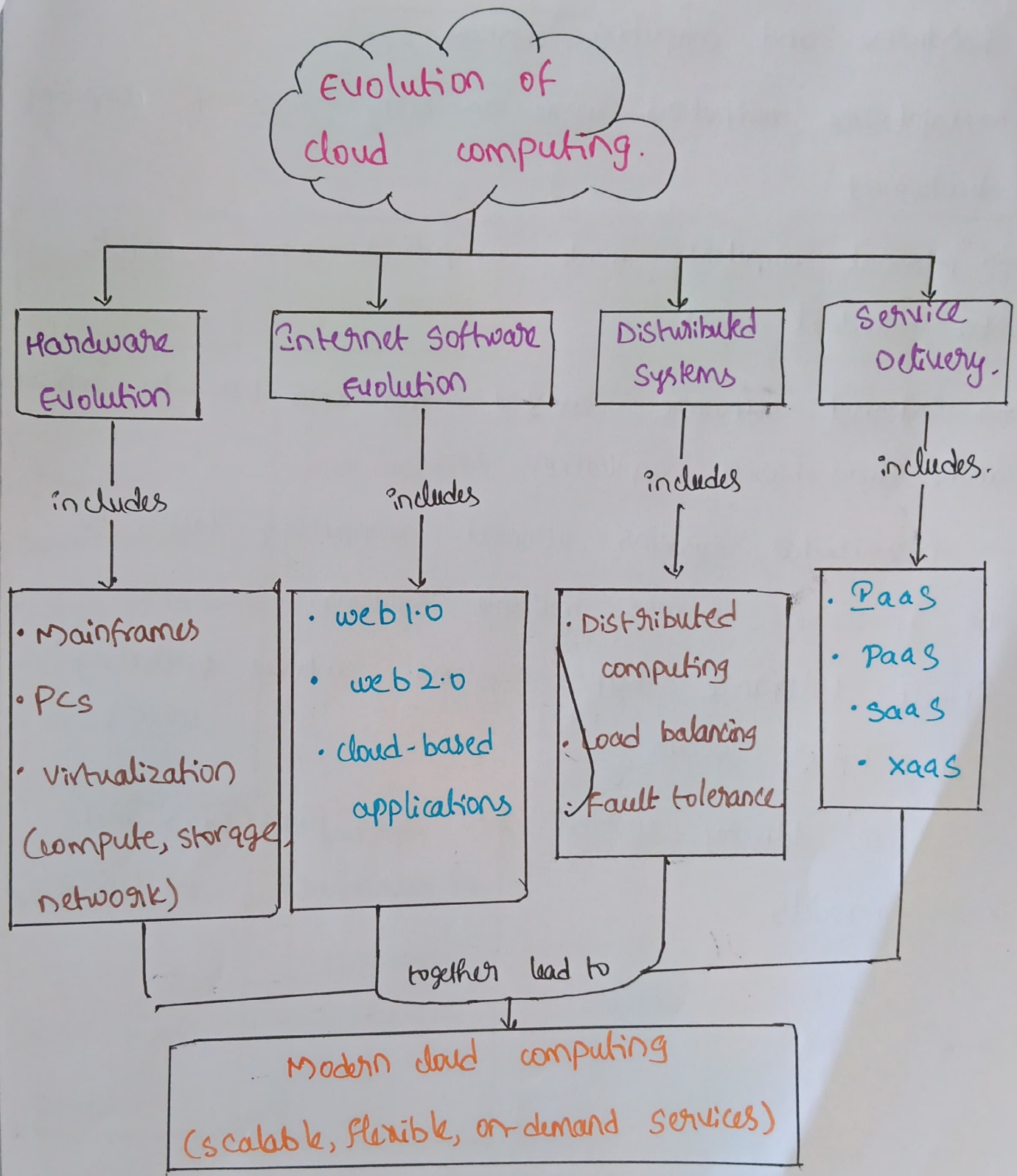
→ cloud services depend on internet connectivity

→ users have limited control over the underlying Infrastructure.

Evolution

Resources

Evolution of cloud computing.



→ cloud computing evolved from the development hardware and computing systems.

→ Mainframe computers were an early stage of computer development.

→ personal computers made computing more accessible to individual users.

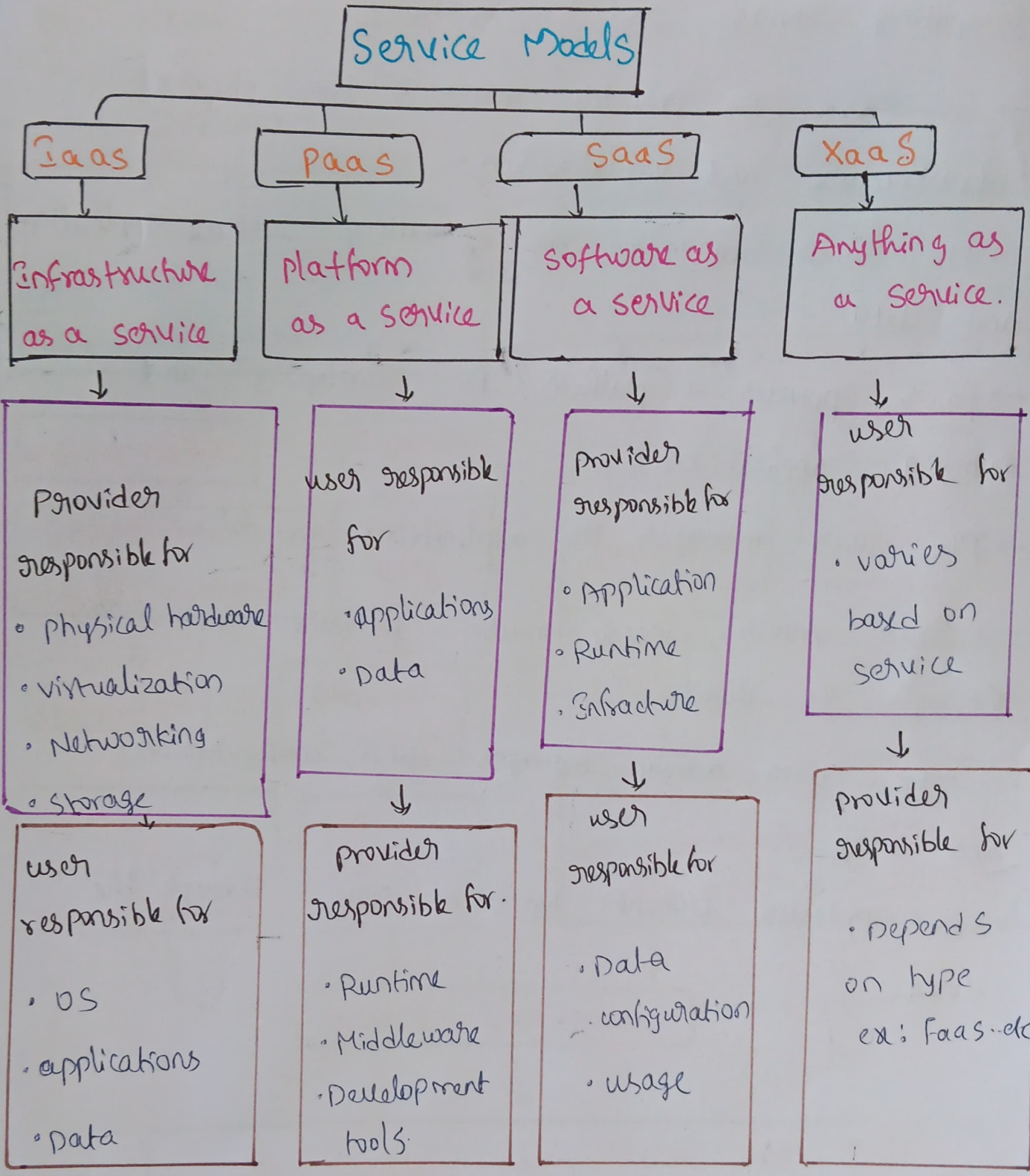
→ Internet software evolved from web 1.0 to web 2.0 and cloud-based applications.

→ Distributed systems allowed computing tasks to be processed across multiple systems.

→ Fault tolerance helps systems continue operating when failure occur.

→ Service delivery evolved into Iaas, Paas, Saas and Xaas models.

Service Models.



→ Iaas provides infrastructure resources such as computing, storage and networking.

→ In Iaas, the provider manages the physical infrastructure and virtualization.

→ The user manages the operating system, applications and data.

→ paas provides a platform for developing and running applications.

→ The user manages the applications and data

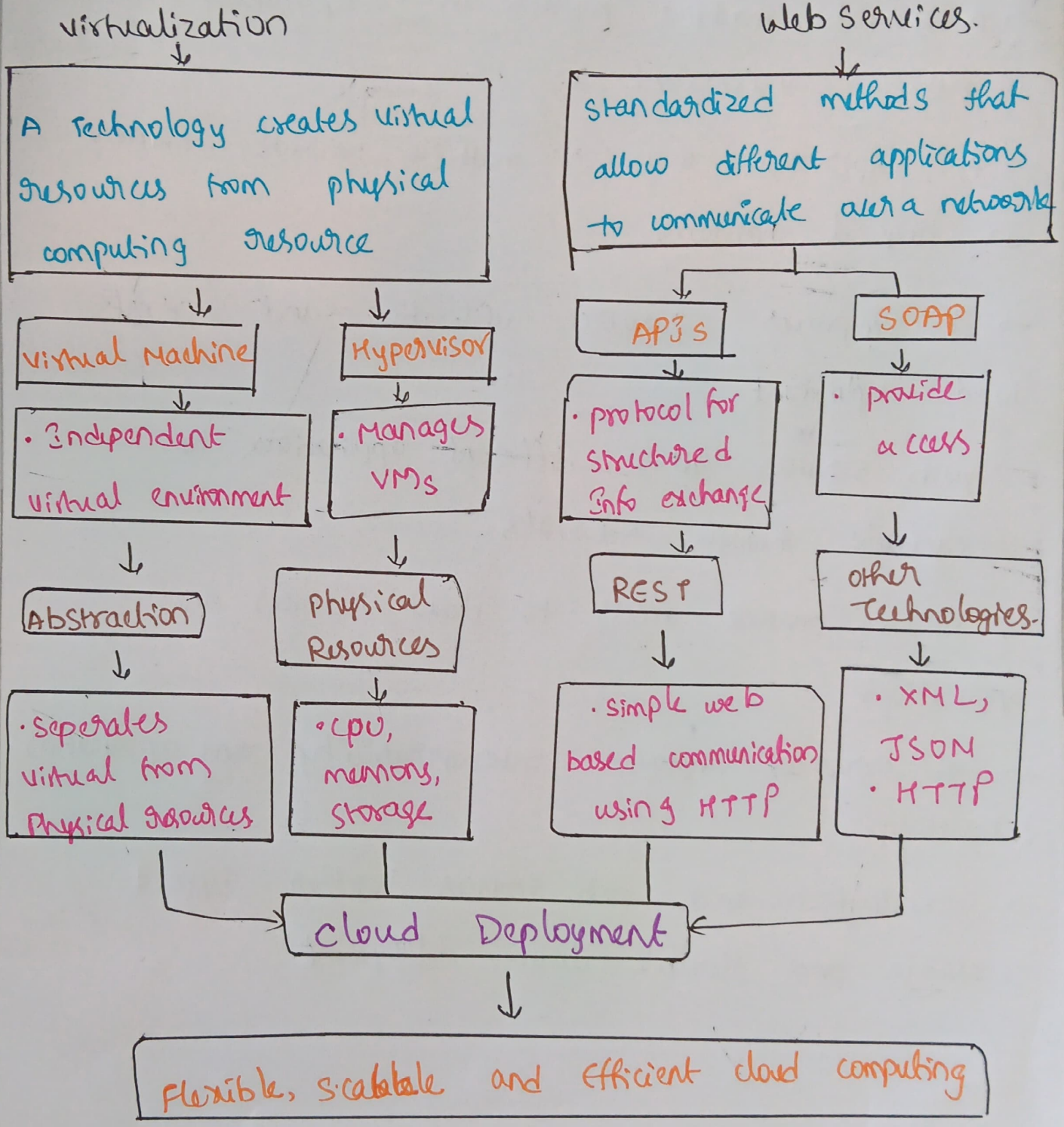
→ Saas provides ready-to-use software applications through the cloud.

→ The user mainly manages data, configuration and usage.

→ xaas provides different types of services through the cloud.

as

virtualization and web services as Enablers of cloud.



→ Virtualization creates virtual resources from physical computing resources.

→ A virtual machine provides an independent virtual computing environment.

→ A hypervisor manages multiple virtual machines on physical hardware.

→ It improves resource utilization and supports cloud deployment.

→ web services allow different applications to communicate through networks.

→ APIs provide access to cloud services and applications

→ web services improves interoperability and application integration.

→ virtualization and web services together support scalable and flexible cloud deployment

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented